Drive systems & mechanisms

Higher · Mechanisms & structures · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Describe one way to reduce wear.

- Lubricate with oil or grease to reduce friction
- Use bearings
- Use harder or self-lubricating materials

3 Quick check

- a) 3:1
- b) No different
- c) 10 N m
- d) Reciprocating motion

4 Key terms

- 1 – E
- 2 – D
- 3 – B
- 4 – C
- 5 – A
- 6 – F

5 Calculate the power output.

- $\omega = 2\pi N/60 = 2\pi \times 1200 \div 60 = 125.7$ rad/s
- $P = T\omega = 2.4 \times 125.7 = 302$ W (300 W)

6 Calculate the torque on the drum.

- $F = mg = 50 \times 9.8 = 490$ N
- $T = F \times r = 490 \times 0.1 = 49$ N m

No marks for: Using the diameter instead of the radius

7 Calculate the velocity ratio and the output speed.

- VR = driven ÷ driver = 60 ÷ 20 = 3
- Output speed = 1500 ÷ 3
- = 500 rpm